Astrodynamics 2

Equation 8-15

2020.11.19

김나은

Show that the equation 8-15

- Equation 8-15

$$P_l[\cos(\Lambda)] = P_l[\sin(\varphi_{gcQ})]\, P_l[\sin(\varphi_{gcsat})] + 2\sum_{m=1}^{l}\frac{(l-m)!}{(l+m)!}\, A_{l,m}A'_{l,m} + B_{l,m}B'_{l,m}$$

$$where\ A_{l,m} = P_{l,m}[\sin(\varphi_{gcQ})]\cos(m\lambda_Q)$$

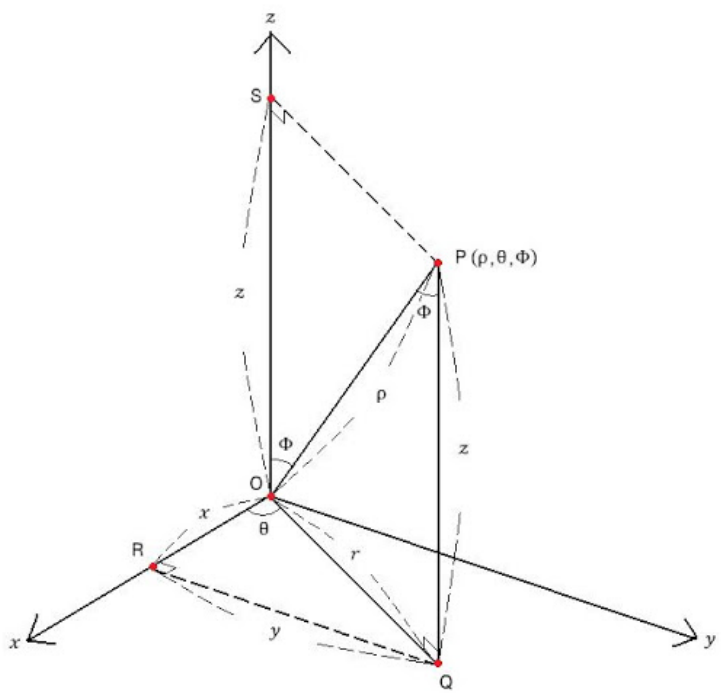
$$A'_{l,m} = P_{l,m}[\sin(\varphi_{gcsat})]\cos(m\lambda_{sat})$$

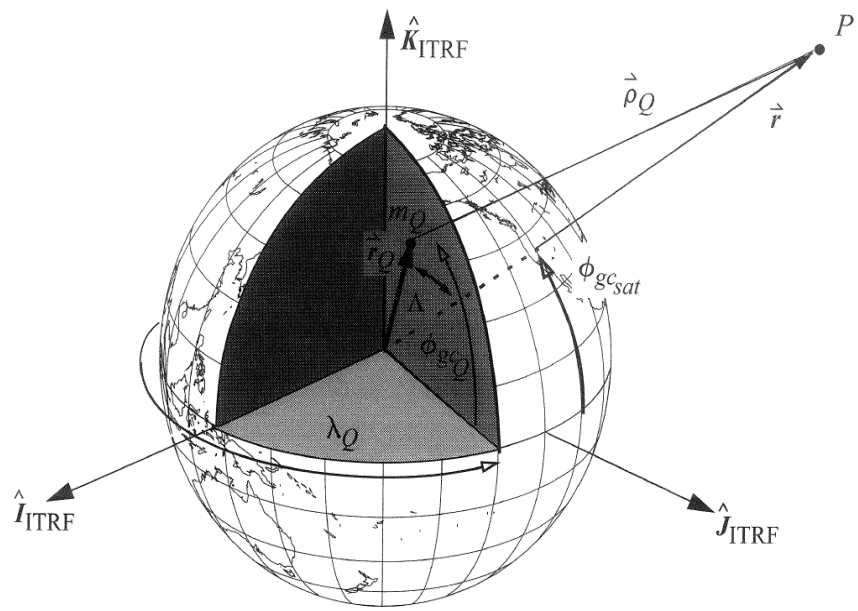
$$B_{l,m} = P_{l,m}[\sin(\varphi_{gcQ})]\sin(m\lambda_Q)$$

$$B'_{l,m} = P_{l,m}[\sin(\varphi_{gcsat})]\sin(m\lambda_{sat})$$

Show that the equation 8-15

- Spherical harmonics
  - 구의 표면에서의 임의의 function, f(φ, θ)은 spherical harmonics, (φ, θ)에 관한 polynomials $Y_l^m(\varphi,\theta)$ 의 linear sum으로 나타낼 수 있다.
  - $Y_l^m(\varphi,\theta)$ (구면조화함수) = $\sqrt{\frac{2l+1}{4\pi}\frac{(n-m)!}{(n+m)!}}P_l^m(\cos\varphi)\,e^{im\theta}$





Show that the equation 8-15

- Spherical harmonics

Let ($\theta \rightarrow \lambda$)

$$f(\varphi,\lambda) = \sum_{l=0}^{\infty}\sum_{m=-l}^{l} A_{l,m}\ P_{l,m}(\cos\varphi)e^{im\lambda} = \sum_{l}^{\infty}\sum_{m=-l}^{l} B_{l,m} Y_{l,m}\ (\varphi, \lambda)$$

where $P_{l,m}(= P_l^{\ m})$ is associated Legendre polynomial,

the solution of the equation, $(1-x^2)y'' - 2xy' + [\ n(n+1) - \frac{m^2}{1-x^2}]\ y = 0$

where $Y_{l,m}\ (\varphi,\lambda) = \sqrt{\frac{2l+1}{4\pi}\frac{(n-m)!}{(n+m)!}} P_l^{\ m}(\cos\varphi)\, e^{im\lambda}$

$$B_{l,m} = \int Y_{l,m}^{\ *}(\Omega) f(\Omega)\ d\Omega$$

$Y_{l,m}$: orthogonal i.e. $\int Y^{*l,m}\ (\Omega) Y_{l',m'}\ (\Omega)\ d\Omega = \delta_{l,m}\ \delta_{l',m'}$

$$Y_{l,m}^{\ *} = (-1)^m Y_{l,-m}$$

Show that the equation 8-15

We can set the equation

$$P_l[\cos\Lambda] = \frac{4\pi}{2l+1}\sum_{m=-l}^{l} Y_{l,m}{}^{*}(\varphi',\lambda')Y_{l,m}(\varphi,\lambda) \quad \text{.....} (1)$$

And

$$Y_{l,m}(\varphi,\lambda) = \sqrt{\frac{2l+1}{4\pi}\frac{(l-m)!}{(l+m)!}}P_l{}^{m}[\cos\varphi]\ e^{im\lambda}$$

$$Y_{l,m}{}^{*}(\varphi,\lambda) = (-1)^m Y_{l,-m}(\varphi,\lambda)$$

Put $Y_{l,m}(\varphi,\lambda)$, $Y_{l,m}{}^{*}(\varphi,\lambda)$ to $P_l[\cos\Lambda]$

$$P_l\,[\cos\Lambda] = \frac{4\pi}{2l+1}\sum_{m=-l}^{l}\ [\,(-1)^m\sqrt{\frac{2l+1}{4\pi}\frac{(l+m)!}{(l-m)!}}P_l{}^{-m}[\cos\varphi']\,e^{-im\lambda'}\sqrt{\frac{2l+1}{4\pi}\frac{(l-m)!}{(l+m)!}}P_l{}^{m}[\cos\varphi]\,e^{im\lambda}\ ]$$

Show that the equation 8-15

- l, m에 관계없는 식은 summation 밖으로 꺼내고, exponential term을 정리

$$P_l\,[\cos\Lambda] = \frac{4\pi}{2l+1}\frac{2l+1}{4\pi}\sum_{m=-l}^{l}\left[\,(-1)^m P_l{}^{-m}[\cos\varphi']\;P_l{}^{m}[\cos\varphi]\;e^{im(\lambda-\lambda')}\,\right]$$

$$= \sum_{m=-l}^{l}(-1)^m P_l{}^{-m}[\cos\varphi']\;P_l{}^{m}[\cos\varphi]\;e^{im(\lambda-\lambda')}$$

- $\sum_{m=-l}^{l} = \sum_{m=-l}^{-1} + \sum_{m=0} + \sum_{m=1}^{l}$ 의 세 경우로 나누면,

$$P_l[\cos\Lambda] = \sum_{m=-l}^{-1}(-1)^m P_l{}^{-m}[\cos\varphi']\;P_l{}^{m}[\cos\varphi]\;e^{im(\lambda-\lambda')} + P_l\;[\cos\varphi']\,P_l\;[\cos\varphi]$$

$$+\sum_{m=1}^{l}(-1)^m P_l{}^{-m}[\cos\varphi']\;P_l{}^{m}[\cos\varphi]\;e^{im(\lambda-\lambda')}$$

Show that the equation 8-15

- 첫 번째 항 :

$$\sum_{m=-l}^{-1} (-1)^m P_l^{-m}[\cos\varphi'] \, P_l^{m}[\cos\varphi] \, e^{im(\lambda-\lambda')} = \sum_{m=1}^{l} (-1)^{(-m)} P_l^{-(-m)}[\cos\varphi'] \, P_l^{-m}[\cos\varphi] \, e^{i(-m)(\lambda-\lambda')}$$

$$= \sum_{m=1}^{l} (-1)^m P_l^{m}[\cos\varphi'] \, P_l^{-m}[\cos\varphi] \, e^{im(-(\lambda-\lambda'))}$$

- Since $P_l^{-m} = (-1)^m \frac{(l-m)!}{(l+m)!} P_l^{m}$

$$\sum_{m=1}^{l} (-1)^m P_l^{m}[\cos\varphi'] \, P_l^{-m}[\cos\varphi] \, e^{im(-(\lambda-\lambda'))} = \sum_{m=1}^{l} (-1)^m P_l^{m}[\cos\varphi'] \, (-1)^m \frac{(l-m)!}{(l+m)!} P_l^{m}[\cos\varphi] \, e^{im\left(-(\lambda-\lambda')\right)}$$

$$= \sum_{m=1}^{l} \frac{(l-m)!}{(l+m)!} P_l^{m}[\cos\varphi'] \, P_l^{m}[\cos\varphi] \, e^{im(-(\lambda-\lambda'))}$$

Show that the equation 8-15

- 세 번째 항 :

$$\sum_{m=1}^{l} (-1)^m P_l^{-m}[\cos\varphi'] \; P_l^{m}[\cos\varphi] \; e^{im(\lambda-\lambda')} = \sum_{m=1}^{l} (-1)^m \; (-1)^m \; \frac{(l-m)!}{(l+m)!} P_l^{m}[\cos\varphi'] \; P_l^{m}[\cos\varphi] \; e^{im(\lambda-\lambda')}$$

$$= \sum_{m=1}^{l} \frac{(l-m)!}{(l+m)!} P_l^{m}[\cos\varphi'] \; P_l^{m}[\cos\varphi] \; e^{im(\lambda-\lambda')}$$

- Therefore,

$$P_l[\cos\Lambda] = \sum_{m=1}^{l} \frac{(l-m)!}{(l+m)!} P_l^{m}[\cos\varphi'] \; P_l^{m}[\cos\varphi] \; e^{im\left(-(\lambda-\lambda')\right)} + P_l[\cos\varphi'] \; P_l[\cos\varphi]$$

$$+ \sum_{m=1}^{l} \frac{(l-m)!}{(l+m)!} P_l^{m}[\cos\varphi'] \; P_l^{m}[\cos\varphi] \; e^{im(\lambda-\lambda')}$$

$$= P_l[\cos\varphi']P_l[\cos\varphi] + \sum_{m=1}^{l} \frac{(l-m)!}{(l+m)!} P_l^{m}[\cos\varphi'] \; P_l^{m}[\cos\varphi] \left(e^{im(\lambda-\lambda')} + e^{im(-(\lambda-\lambda'))} \right)$$

Show that the equation 8-15

- By Euler formula, $e^{im(\lambda-\lambda')} + e^{im(-(\lambda-\lambda'))} = 2\cos(\lambda-\lambda')$

$$P_l[\cos\Lambda] = P_l[\cos\varphi']P_l[\cos\varphi] + \sum_{m=1}^{l} \frac{(l-m)!}{(l+m)!} P_l^{\,m}[\cos\varphi'] \; P_l^{\,m}[\cos\varphi] \, (2\cos(\lambda-\lambda'))$$

$$= P_l[\cos\varphi]P_l[\cos\varphi'] + 2\sum_{m=1}^{l} \frac{(l-m)!}{(l+m)!} P_l^{\,m}[\cos\varphi] \; P_l^{\,m}[\cos\varphi'] \, (\cos(\lambda-\lambda'))$$

Show that the equation 8-15

- General spherical coordinate to topocentric horizon (SEZ)

$$\varphi = 90 - \varphi_{gcQ}$$

$$\varphi' = 90 - \varphi_{gcsat}$$

$$\lambda = \lambda_Q$$

$$\lambda' = \lambda_{sat}$$

- Then

$$\cos(\lambda - \lambda') = \cos\lambda \cos\lambda' + \sin\lambda \sin\lambda' \qquad (\because \cos(\alpha - \beta) = \cos\alpha \cos\beta + \sin\alpha \sin\beta)$$

$$= \cos\lambda_Q \cos\lambda_{sat} + \sin\lambda_Q \sin\lambda_{sat}$$

Show that the equation 8-15

$$\therefore \quad P_l\ [\cos\Lambda] = P_l[\sin\varphi_{gcQ}]P_l[\sin\varphi_{gcsat}] +$$

$$2\sum_{m=1}^{l} \frac{(l-m)!}{(l+m)!} P_l^{\,m}[\sin\varphi_{gcQ}]P_l^{\,m}[\sin\varphi_{gcsat}]\,(\cos\lambda_Q \cos\lambda_{sat} + \sin\lambda_Q \sin\lambda_{sat})$$

$$= P_l\,[\sin(\varphi_{gcQ})]\ P_l\,[\sin(\varphi_{gcsat})] + 2\sum_{m=1}^{l} \frac{(l-m)!}{(l+m)!} A_{l,m}A'_{l,m} + B_{l,m}B'_{l,m}$$

$$A_{l,m} = P_{l,m}[\sin(\varphi_{gcQ})]\cos(m\lambda_Q)$$

$$A'_{l,m} = P_{l,m}[\sin(\varphi_{gcsat})]\cos(m\lambda_{sat})$$

$$B_{l,m} = P_{l,m}[\sin(\varphi_{gcQ})]\sin(m\lambda_Q)$$

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식 (1) 유도

- Coordinate Transformation

For an arbitrary function $g(\varphi, \lambda)$,

$$g(\varphi,\lambda) = Y_{l,m}(\varphi,\lambda) = \sum_{m=-l}^{l} a_{l,m}\, Y_{l,m}(\Lambda,\beta) \text{ in } (x', y', z').$$

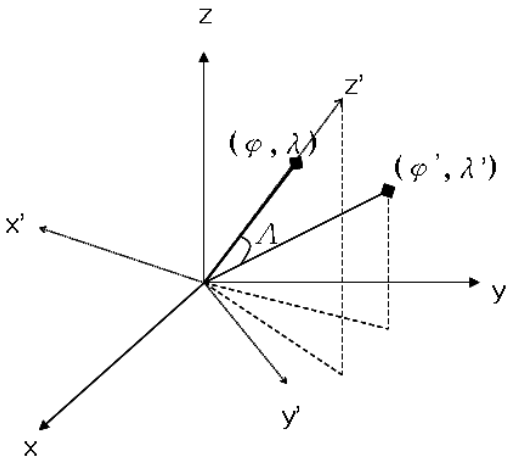
$$where\ a_{l,m} = \int g(\varphi,\lambda)\, Y_{l,m}{}^{*}(\Lambda,\beta)\, d\Omega_{\varphi,\lambda}$$

- Then, we can express the function

$$f(\Lambda,\beta) = P_l(\cos\Lambda) = \sum_{m=-l}^{l} b_{l,m}\, Y_{l,m}(\varphi,\lambda) \text{ in } (x,y,z).$$

$$where\ b_{l,m} = \int f(\Lambda,\beta)\, Y_{l,m}{}^{*}(\varphi,\lambda)\, d\Omega_{\Lambda,\beta} \ \ldots\ldots (*)$$

=> $P_l[\cos\Lambda]$을 얻기 위해 $b_{l,m}$ 을 구하자!!



식 (1) 유도

Consider the case of $\Lambda = 0$.

$$\rightarrow g(\varphi,\lambda)\,|_{\Lambda=0} \;=\; \sum_{m=-l}^{l} a_{l,m}\, Y_{l,m}(\Lambda,\beta)|_{\Lambda=0}$$

$$= \sum_{m=-l}^{l} a_{l,m}\,(-1)^m \sqrt{\frac{(2l+1)}{4\pi}\frac{(l-m)!}{(l+m)!}}\, p_l{}^m(1) e^{im\lambda} = a_{l,0}\sqrt{\frac{(2l+1)}{4\pi}}$$

Since $m \neq 0 \;\rightarrow\; P_l{}^m(1) \;=\; 0,$

$m \;=\; 0 \;\rightarrow\; P_l(1) \;=\; 1$

$$\textit{where}\;\; a_{l,0} \;=\; \int g(\varphi,\lambda) Y_{l,0}{}^*(\Lambda,\beta)\; d\Omega \;=\; \int Y_{l,m}(\varphi,\lambda)\; Y_{l,0}{}^*(\Lambda,\beta)\; d\Omega$$

식 (1) 유도

- 식 (*)에 의해 $b_{l,m} = \int \quad P_l\,[\cos\Lambda]\; Y_{l,m}{}^{*}(\varphi,\lambda)\; d\Omega_{\Lambda,\beta}$ 로 나타내어진다.

If m=0,

$$Y_{l,0}\,(\Lambda,\beta) \;=\; \sqrt{\frac{(2l+1)}{4\pi}}\,P_l[\cos\Lambda] \;\Rightarrow\; P_l[\cos\Lambda] \;=\; \sqrt{\frac{4\pi}{2l+1}}\,Y_{l,0}\,(\Lambda,\beta)$$

Then, $b_{l,m} = \int \quad \sqrt{\frac{4\pi}{(2l+1)}}\,Y_{l,0}(\Lambda,\beta)\; Y_{l,m}{}^{*}(\varphi,\lambda)\; d\Omega_{\Lambda,\beta}$

식 (1) 유도

- Conjugate를 양변에!

$$b_{l,m}{}^{*} = \int \sqrt{\frac{4\pi}{(2l+1)}} Y_{l,0}{}^{*}(\Lambda,\beta)\ Y_{l,m}\ (\varphi,\lambda)\ d\Omega_{\Lambda,\beta} = \sqrt{\frac{4\pi}{(2l+1)}} \int Y_{l,0}{}^{*}(\Lambda,\beta)\ Y_{l,m}\ (\varphi,\lambda)\ d\Omega_{\Lambda,\beta}$$

$\int Y_{l,m}(\varphi,\lambda)\ Y_{l,0}{}^{*}(\Lambda,\beta)\ d\Omega = a_{l,0}$ 이고, $a_{l,0} = \sqrt{\frac{4\pi}{2l+1}} g(\varphi,\lambda)\,|_{\Lambda=0}$ 이므로,

- $b_{l,m}{}^{*} = \sqrt{\frac{4\pi}{(2l+1)}}\sqrt{\frac{4\pi}{(2l+1)}}\ g(\varphi,\lambda)|_{\Lambda=0} = \frac{4\pi}{(2l+1)}\ Y_{l,m}\ (\varphi,\lambda)|_{\Lambda=0} = \frac{4\pi}{(2l+1)}\ Y_{l,m}\ (\varphi',\lambda')$

$$\therefore\ b_{l,m} = \frac{4\pi}{(2l+1)}\ Y_{l,m}{}^{*}(\varphi',\lambda')$$

식 (1) 유도

$$P_l[\cos\Lambda] = \sum_{m=-l}^{l} b_{l,m}\, Y_{l,m}(\varphi,\lambda)$$ 이므로,

$$\therefore \quad P_l[\cos\Lambda] = \frac{4\pi}{2l+1}\sum_{m=-l}^{l} Y_{l,m}{}^{*}(\varphi',\lambda')Y_{l,m}(\varphi,\lambda)$$

Thank you for listening